Radon FTLab

FRD500 Radon in Soil Quick Manual (User's Manual)

Radon FTLab



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※ This device is an electromagnetic compatible device for business use(Class A)that the seller or the user should be aware of, and is intended for use besides home.

FRD500 Radon in Soil Kit Quick Manual





Photo [1]

➢ How to Install Probe

1. Assemble the Retract-A-Tip in the Probe Box referring to the bottom part of the Photo [1] and connect the hose connector to the inside of the Tip. Cut an opaque 2m-long hose to the hose connector. Then, connect as shown the top part of the Photo [1]. Now the Probe preparation is done.
2. Drive the prepared probe into the ground using a slide hammer. Drive into the ground the minimum of 30cm. 50cm depth is recommended.
3. Slightly pulling the probe stuck in the ground will cause Retract-A-Tip to be lifted. Now it's ready for measurement. Finally, stomp the ground around Probe to prevent air from entering the gap as much as possible.

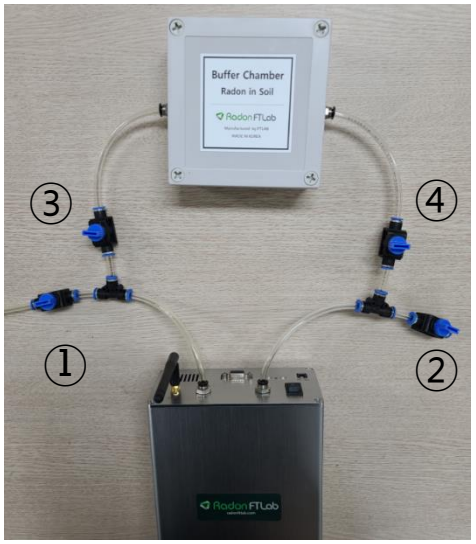
➢ How to Measure

1. Connect the Antenna to the Antenna Port on the rear of the FRD500.
2. Connect the DC 5V/2A adaptor provided with the DC 5V power connector or the auxiliary battery having a higher power, and turn on the Power Switch on the back.
3. Select Radon in Water mode using Button on the measurement mode selection screen. If it is disabled in dark gray, connect the smartphone app to activate Radon in Water in Config Setting on the upper right side. It can also be activated in the Parameter Setup in the PC program.
4. Connect the opaque hose stuck in the ground and the transparent hose cut into 40cm with the connector on the Soil Kit Box. Connect the end of the transparent hose to the inlet on the rear of the FRD500 with the enclosed PTFE filter.
5. After checking that there are no leaks in the hose connection, press Start/Stop Button to start the measurement. The measurement order is; 10 minutes pumping - 5 minutes waiting - 10 minutes measurement 4 times, with the total of 55 minutes. By the third measurement, the display will show the result, and after the fourth measurement, the average of the 4 measurements will appear on the display, and the measurement will be automatically terminated.
   (Data is automatically saved when the measurement is complete.)
6. The measured value is 1/6 of the value displayed on the FRD500 Display. For example, if the displayed value were 1,800 pCi/L, the actual measurement is 300 pCi/L.
7. Since Radon in Water mode has a maximum measurement range of 15,000 pCi/L, the maximum measurement value is 2,500 pCi/L. To measure a soil at a higher concentration, connect the components as shown in Photo [3].
8. Open Valve ① and Valve ② and close Valve ③ and Valve ④, and operate for 10 minutes in the Flushing mode.
9. After 10 minutes of operation in Flushing mode, close Valve ① and Valve ② and open Valve ③ and Valve ④ to make a closed loop including Buffer Chamber, and start measurements in Radon in Water mode.
10. When the Buffer Chamber is used, the measured value is one-half of the value displayed on the FRD500 Display. For example, if the displayed value is 1,800 pCi/L, the actual measurement is 900 pCi/L.
11. This time also, Since Radon in Water mode has a maximum measurement range of 15,000 pCi/L, the maximum measurement value is 7,500 pCi/L.

* Since soil radon is measured using the water measurement mode, the displayed value is the programmed to suit the water measurement mode. So, when there is no buffer chamber apply by calculating by 1/6, and 1/2 when using buffer chamber, by 1/2. (It is recommended to use buffer chamber)



[Photo 2] Radon in Soil Measurement



[Photo 3] Buffer chamber Connection

➢ Data Save & Check

1. The stored Log is marked with Stored Time and Radon in Water in the Log List.
2. The log consists of 4 measurements and an average value. You can check the measured value on the FRD500 device itself, or you can also use a smartphone app or PC program.
3. Use one Slot for a single measurement for the Log of Radon in Water. If there are 10 stored logs, the oldest log will be automatically deleted, so the data must be backed up.
4. You can check and save it in the app or PC program even during measurement.
   (Real-time measurements are displayed when Auto is set in PC program)

➢ Cautions

1. Make sure to check the directions of the Inlet and Outlet so that accurate measurements can be made.
2. If you use an adaptor and battery with a power of 5V/2A or less, pumping and measurement may not proceed normally.
3. Without Buffer Chamber, measurements can be made up to 92,500 Bq/m3 (2,500 pCi/L) (1/6), with Buffer Chamber, up to 277,500 Bq/m3 (7500 pCi/L)..
4. The PTFE filter must be connected and used to prevent soil and dust from entering the FRD500 during measurement.
5. Before starting the measurement, operate the Buffer Chamber and the FRD500 in the well ventilated area for at least 10 minutes in the Flushing mode. In particular, it is strongly recommended to flush for at least 20 minutes after measuring high concentrations of radon. When measuring again immediately after measuring a high concentration of radon, Flushing mode must be used, as the remaining radon at the Buffer Chamber and the measurement section interfere with the new measurement.